
SPIRITUAL WELLNESS & MEDITATION APP

Comprehensive Project Implementation Plan

EDAI 6 - Group Project

Team

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1. Executive Summary & Project Vision

The MAM Spiritual Wellness & Meditation Application is a cross-platform mobile application designed to make meditation, mindfulness, and spiritual wellness practices accessible to a broad audience. The app targets users ranging from absolute beginners who have never meditated to experienced practitioners looking for structured courses and community engagement.

1.1 Problem Statement

Despite the growing awareness of mental health and wellness, there is a significant gap in the Indian and South Asian market for a spiritual wellness app that combines traditional meditation practices with modern habit-tracking technology. Existing global apps like Calm and Headspace are primarily Western-centric in their approach and pricing. Our application fills this gap by offering culturally relevant content, affordable subscription tiers, and a holistic approach that includes yoga, pranayama, chanting, and guided meditation.

1.2 Vision Statement

To become the leading spiritual wellness platform in the Indian subcontinent, empowering 1 million users within the first two years to build sustainable meditation habits through technology-driven, culturally relevant content.

1.3 Key Objectives

- Deliver a smooth, performant cross-platform mobile experience on both iOS and Android from a single codebase.
- Provide a rich content library of guided meditations, video lessons, podcasts, and daily spiritual quotes.
- Implement a robust habit-tracking and streak system to drive daily engagement and long-term retention.
- Enable live events and community engagement features for a sense of belonging.
- Build a scalable admin panel for content management, analytics, and user administration.
- Achieve a Day-7 retention rate of at least 40% and Day-30 retention of at least 20%.

1.4 Success Metrics

Metric	Target (6 months)	Target (12 months)
Monthly Active Users (MAU)	25,000	100,000
Day-1 Retention	60%	65%
Day-7 Retention	35%	40%
Day-30 Retention	15%	20%
Trial-to-Paid Conversion	5%	8%
Average Session Duration	8 minutes	12 minutes

Metric	Target (6 months)	Target (12 months)
App Store Rating	4.3+	4.5+
Crash-Free Sessions	99.5%	99.8%

2. Market Analysis & Competitive Landscape

2.1 Market Size & Opportunity

The global meditation apps market generated approximately \$1.6 billion in revenue in 2024 and is projected to reach around \$2.25 billion in 2025, growing at roughly 18% CAGR. The broader health and wellness industry is expected to grow from \$6.87 trillion in 2025 to approximately \$11 trillion by 2034. Within India specifically, the digital wellness market is nascent but rapidly expanding, driven by increasing smartphone penetration, 5G rollout, and growing mental health awareness among the 18-35 demographic.

2.2 Competitive Analysis

App	Strengths	Weaknesses	Our Differentiator
Calm	Premium content, celebrity narrators, sleep stories	Expensive (\$70/yr), Western-focused	Culturally relevant Indian spiritual content at lower price
Headspace	Strong branding, structured courses, animations	Limited free content, no live events	Live spiritual events, generous free tier
Insight Timer	Large free library, community features	Overwhelming UI, inconsistent quality	Curated quality content, clean minimal UI
BrahmaKumar is App	Strong spiritual content, Indian focus	Dated UI, no habit tracking, no streaks	Modern UX with gamification and tracking

2.3 Target Audience Segments

- **Segment A - Curious Beginners (18-28):** College students and young professionals exploring meditation for the first time. Need short sessions (3-5 min), gamification, and social proof.
- **Segment B - Stressed Professionals (25-40):** Working professionals seeking stress relief and sleep improvement. Willing to pay for premium content, value progress tracking.
- **Segment C - Spiritual Practitioners (30-55):** Users with existing meditation or yoga practice looking for advanced courses, pranayama, and community events.
- **Segment D - Wellness Seekers (35-60):** Health-conscious individuals integrating meditation into a broader wellness routine. Interested in holistic content including Ayurveda and yoga.

3. System Architecture & Technical Design

3.1 High-Level Architecture Overview

The application follows a three-tier architecture with clear separation of concerns: a Presentation Layer (mobile app and admin panel), a Business Logic Layer (backend API with middleware), and a Data & Storage Layer (database and media storage). This is supplemented by an External Integration Layer for third-party services.

3.2 Architecture Layers

Presentation Layer

- **Mobile App (React Native CLI):** The primary user-facing application built with React Native using the bare React Native CLI workflow for full native control, custom native modules, and direct access to native build tools (Xcode and Android Studio).
- **Admin Panel (React.js / Web):** A web-based dashboard built with React.js for content management, user administration, event scheduling, analytics viewing, and notification dispatch.

Business Logic Layer

- **Node.js & Express Backend:** RESTful API server handling authentication flows, business logic, data validation, and orchestration between services.
- **Global Exception Handler (Middleware):** Centralized error handling that logs error details, maps error codes to user-friendly messages, and sends generic responses to prevent information leakage.
- **Core Logic Modules:** Auth Logic (OTP-based authentication), Habit Tracking Logic (streak calculations, daily logging), Payment Logic (subscription management), and Course Orchestration (enrollment, progress tracking, content delivery).

Data & Storage Layer

- **PostgreSQL (via Supabase):** Primary relational database storing user profiles, habit logs, course metadata, event records, subscription data, and analytics events.
- **Cloudflare R2 / AWS S3 (Media Storage):** Object storage for video lessons, podcast audio files, thumbnail images, and other media assets. Cloudflare Stream for adaptive bitrate video delivery.

External Integration Layer

- **OTP Service:** SMS/WhatsApp OTP delivery for phone-based authentication (e.g., MSG91 or Twilio).
- **Payment Gateway:** Razorpay for Indian payments (UPI, cards, wallets) and Stripe for international payments.
- **Push Notifications:** Firebase Cloud Messaging (FCM) for both iOS and Android push notifications.
- **Analytics:** Mixpanel for user behavior analytics, funnel analysis, and retention tracking.

3.3 Communication Protocols

All communication between the mobile app and backend uses JSON over HTTPS (TLS 1.3). The admin panel communicates with the same backend API using identical protocols. Real-time features such as live event streaming use WebSocket connections. Media assets are served via CDN with signed URLs for

access control.

3.4 Data Flow Summary

Flow	Source	Destination	Protocol	Data
User Login	Mobile App	Backend API	HTTPS POST	Phone number, OTP
Load Home Feed	Mobile App	Backend API + CDN	HTTPS GET	Videos, quotes, events
Start Meditation	Mobile App	Local + Backend	Local / HTTPS POST	Session timer, duration
Enroll Course	Mobile App	Backend API	HTTPS POST	Course ID, user token
Process Payment	Mobile App	Razorpay + Backend	HTTPS POST	Payment intent, plan ID
Push Notification	Backend API	FCM	HTTPS POST	Notification payload
Admin Content Upload	Admin Panel	Backend + R2/S3	HTTPS POST	Media files, metadata

4. Technology Stack - Deep Dive

Each technology in our stack has been chosen for a specific reason. Below is a detailed justification for every tool and how it fits into the project.

4.1 Frontend - Mobile App

Technology	Version	Purpose	Why This Choice
React Native (CLI)	0.76+	Cross-platform mobile framework	Single codebase for iOS & Android; large ecosystem; hot reloading; full native access
React Navigation	v7	Screen navigation	Stack, tab, drawer navigators; deep linking support; type-safe routes
NativeWind	v4	Tailwind CSS for React Native	Utility-first styling, consistent design tokens, rapid UI development
React Query	v5	Server state management	Caching, background refetching, optimistic updates, offline support
Zustand	v4	Client state management	Lightweight, no boilerplate, middleware support for persistence
React Hook Form + Zod	Latest	Form handling + validation	Performant forms with runtime type-safe validation
react-native-video	v6+	Video playback	HLS streaming, background playback, DRM, lock-screen controls
react-native-track-player	Latest	Audio playback	Background audio, lock-screen controls, queue management, streaming

4.2 Frontend - Admin Panel

Technology	Purpose
React.js (Vite)	Fast SPA for admin dashboard with hot module replacement
Tailwind CSS	Consistent styling with the mobile app's design system
Shadcn/UI	Pre-built accessible components (tables, modals, forms, charts)
TanStack Table	Powerful data tables with sorting, filtering, pagination
Recharts	Analytics dashboard charts and visualizations
React Router v7	Client-side routing with nested layouts

4.3 Backend

Technology	Purpose	Justification
Node.js (v22 LTS)	Runtime environment	Non-blocking I/O ideal for concurrent streaming and real-time features

Technology	Purpose	Justification
Express.js	HTTP framework	Minimal, flexible, massive middleware ecosystem
Supabase	Backend-as-a-Service	Hosted Postgres, built-in Auth, Edge Functions, real-time subscriptions, Row Level Security
Supabase Edge Functions	Serverless functions	Handle webhooks (Razorpay), cron jobs (streak reset), compute-heavy tasks
Supabase Storage	File storage	Integrated with RLS, signed URLs, image transformations
JWT (via Supabase Auth)	Authentication tokens	Stateless, scalable, automatic refresh with Supabase client

4.4 Database & Storage

Technology	Purpose	Details
PostgreSQL (Supabase)	Primary database	User profiles, habit logs, courses, events, subscriptions; RLS policies for security
Cloudflare Stream	Video hosting & delivery	Adaptive bitrate streaming, global CDN, DRM support for premium content
Cloudflare R2	Object storage	S3-compatible, zero egress fees, stores audio files, thumbnails, user avatars
Supabase Realtime	Live data sync	Real-time subscriptions for live event chat, notification badges, admin dashboard

4.5 DevOps & Tooling

Tool	Purpose
GitHub	Version control, code reviews, CI/CD via GitHub Actions
Vercel	Admin panel deployment with automatic preview deploys per PR
Fastlane	Automated iOS and Android builds, code signing, and app store submissions
CodePush (App Center)	Over-the-air JS bundle updates without app store review
Sentry	Real-time crash reporting, error tracking, performance monitoring
Claude & Gemini	AI-assisted development, code generation, debugging, documentation, and API schema planning
Figma	UI/UX design, wireframes, prototyping, user flow mapping, and design handoff

5. Database Schema & Data Model

The database is designed around six core domains: Users, Content (Courses & Lessons), Habits, Events, Payments, and Notifications. All tables use UUID primary keys and include created_at / updated_at timestamps. Row Level Security (RLS) policies ensure users can only access their own data unless explicitly shared.

5.1 Core Tables

Table	Key Columns	Purpose	RLS Policy
users	id, phone, name, avatar_url, role, onboarding_complete	User profiles and preferences	Users read/update own row
courses	id, title, description, thumbnail_url, instructor_id, difficulty, is_premium, order	Course catalog	Public read; admin write
lessons	id, course_id, title, media_url, media_type, duration_seconds, order	Individual lessons within courses	Public read; admin write
enrollments	id, user_id, course_id, progress_pct, current_lesson_id, completed_at	User course enrollments and progress	Users read/update own rows
meditation_sessions	id, user_id, duration_seconds, type, started_at, completed	Individual meditation session logs	Users read/insert own rows
habit_logs	id, user_id, log_date, meditation_done, streak_count	Daily habit check-ins	Users read/insert own rows
events	id, title, description, event_date, stream_url, instructor_id, is_live	Live and scheduled events	Public read; admin write
event_registrations	id, user_id, event_id, registered_at	User event registrations	Users read/insert own rows
subscriptions	id, user_id, plan_type, status, started_at, expires_at, gateway_id	Subscription records	Users read own row; admin read all
payments	id, user_id, subscription_id, amount, currency, gateway_ref, status	Payment transaction log	Users read own; admin read all
notifications	id, user_id, title, body, type, read, sent_at	Push notification log	Users read/update own rows
daily_quotes	id, quote_text, author, date, category	Daily spiritual quotes	Public read; admin write
content_directory	id, title, type, media_url, category, tags, is_premium	Browsable content directory (videos, audio)	Public read; admin write

5.2 Key Relationships

- A **user** can have many **enrollments**, each linking to one **course**.
- A **course** has many **lessons** ordered sequentially.

- A **user** has many **meditation_sessions** and **habit_logs** (one per day).
- A **user** has one active **subscription** and many **payments**.
- An **event** has many **event_registrations** linking to users.

5.3 Streak Calculation Logic

Streaks are calculated server-side using a PostgreSQL function. When a user completes a meditation session, the backend checks if a `habit_log` entry exists for today. If not, it creates one and increments the `streak_count` by comparing the previous day's log. If the previous day has no log, the streak resets to 1. A nightly cron job (Supabase Edge Function) handles edge cases like timezone boundaries and ensures streak data integrity.

6. Feature Breakdown & Module Specifications

6.1 Authentication Module

Phone-based OTP authentication flow powered by Supabase Auth with a third-party OTP delivery service.

- **Register/Login:** User enters phone number, receives OTP via SMS, enters OTP, account is created or session is restored.
- **Session Management:** JWT tokens stored securely using react-native-keychain (Keychain on iOS, Keystore on Android); automatic token refresh via Supabase client.
- **Onboarding Flow:** After first login, users go through a 3-screen onboarding: select interests, set daily meditation goal, enable notifications.

6.2 Home Screen

- **Daily Quote:** A rotating spiritual quote displayed prominently at the top.
- **Quick Start Meditation:** One-tap button to start a timed meditation session (with preset durations: 3, 5, 10, 15, 20 min).
- **Trending Videos:** Horizontally scrollable carousel of popular/new video content.
- **Live Events Banner:** If a live event is ongoing or upcoming within 1 hour, show a prominent banner.
- **Podcasts Section:** Featured podcast episodes with play button.
- **Streak Widget:** Current streak count, weekly calendar showing completed days, motivational message.

6.3 Courses Module

- **Browse Courses:** Grid/list view with filters (difficulty, category, duration, free/premium).
- **Course Detail Page:** Description, instructor info, lesson list with duration, user reviews, enroll button.
- **Lesson Player:** Full-screen video/audio player with progress bar, playback speed control, background audio support.
- **Progress Tracking:** Percentage completion per course, resume from last watched lesson, completion certificate.

6.4 My Journey (Habit Tracking)

- **Meditation Logger:** Timer-based meditation session with ambient sound options. Logs duration, type, and timestamp.
- **Streak Dashboard:** Current streak, longest streak, total sessions, total minutes meditated, weekly/monthly heatmap.
- **Daily Check-in:** Simple 'Did you meditate today?' prompt with mood selector.
- **Progress Insights:** Weekly summary with trends, gentle nudges if streak is at risk.

6.5 Content Directory

- **Searchable Library:** Full-text search across all videos, audio, and articles with filters by category and tags.

- **Media Player:** Unified player supporting both video and audio with mini-player for background listening.
- **Favorites & Downloads:** Save content for offline access (premium feature).

6.6 Events Module

- **Event Listing:** Upcoming events with date, time, instructor, description, and registration button.
- **Event Registration:** One-tap registration with calendar reminder integration.
- **Live Streaming:** In-app video streaming for live meditation sessions, workshops, and talks.
- **Event Replay:** Recorded events available for later viewing (premium feature).

6.7 Profile & Settings

- **User Profile:** Name, avatar, stats summary (total sessions, total hours, longest streak).
- **Membership Management:** View current plan, upgrade/downgrade, billing history.
- **Notification Preferences:** Toggle daily reminders, event alerts, new content notifications.
- **App Settings:** Theme (light/dark), meditation timer sounds, language preferences.

6.8 Payments & Subscriptions

- **Subscription Plans:** Free tier, Monthly Premium, Annual Premium (with discount).
- **Payment Flow:** In-app payment via Razorpay (India) or Stripe (international) with UPI, card, and wallet support.
- **Subscription Status:** Active/expired/grace period handling with server-side validation.

6.9 Admin Panel Features

- **User Management:** View, search, filter users; view individual user details and activity; suspend/ban accounts.
- **Content Management:** Upload, edit, delete videos, audio, quotes; organize into courses and categories.
- **Course Management:** Create courses, add/reorder lessons, set pricing (free/premium), publish/unpublish.
- **Event Management:** Create events, set schedules, manage live stream URLs, view registrations.
- **Notification Dispatch:** Send targeted or broadcast push notifications with scheduling.
- **Analytics Dashboard:** DAU/MAU, retention curves, content engagement, revenue, subscription metrics via Mixpanel integration.

7. API Design & Endpoint Reference

The backend exposes a RESTful API with consistent response formats. All endpoints require JWT authentication except for the auth endpoints themselves. Responses follow a standard envelope format with status, data, and error fields.

7.1 Core API Endpoints

Method	Endpoint	Description	Auth
POST	/api/auth/request-otp	Send OTP to phone number	No
POST	/api/auth/verify-otp	Verify OTP and get JWT	No
GET	/api/users/me	Get current user profile	Yes
PATCH	/api/users/me	Update profile details	Yes
GET	/api/home/feed	Home screen data (quotes, videos, events)	Yes
GET	/api/courses	List all courses with filters	Yes
GET	/api/courses/:id	Course details with lessons	Yes
POST	/api/courses/:id/enroll	Enroll in a course	Yes
PATCH	/api/enrollments/:id/progress	Update lesson progress	Yes
POST	/api/sessions	Log a meditation session	Yes
GET	/api/habits/streak	Get current streak and stats	Yes
POST	/api/habits/checkin	Daily habit check-in	Yes
GET	/api/events	List upcoming events	Yes
POST	/api/events/:id/register	Register for an event	Yes
GET	/api/events/:id/stream	Get live stream URL	Yes
GET	/api/directory	Browse content directory	Yes
POST	/api/payments/create-order	Create Razorpay order	Yes
POST	/api/payments/verify	Verify payment callback	Yes
GET	/api/subscriptions/status	Check subscription status	Yes
GET	/api/notifications	List user notifications	Yes

7.2 Response Format

All API responses follow a standard JSON envelope: { "success": true/false, "data": { ... }, "error": { "code": "...", "message": "..." }, "meta": { "page": 1, "total": 100 } }. HTTP status codes are used correctly: 200 for success, 201 for created, 400 for validation errors, 401 for unauthenticated, 403 for unauthorized, 404 for not found, 500 for server errors.

8. Authentication & Security Strategy

8.1 Authentication Flow

The app uses phone-based OTP authentication, which is the most familiar and accessible method for Indian users. The flow is: (1) User enters phone number, (2) Backend requests OTP from OTP service (MSG91/Twilio), (3) OTP is sent via SMS, (4) User enters OTP in app, (5) Backend verifies OTP with the service, (6) Supabase Auth creates/fetches user and returns JWT + refresh token, (7) App stores tokens securely using expo-secure-store.

8.2 Security Measures

- **Transport Security:** All API calls use HTTPS with TLS 1.3. Certificate pinning on mobile app.
- **Token Storage:** JWT stored in react-native-keychain (Keychain on iOS, Keystore on Android). Never in AsyncStorage.
- **Row Level Security:** PostgreSQL RLS policies ensure database-level access control. Users can only query their own data.
- **Rate Limiting:** OTP requests limited to 3 per phone per 10 minutes. API rate limiting at 100 requests/minute per user.
- **Input Validation:** All inputs validated with Zod schemas on both client and server side.
- **Media Access Control:** Premium content served via signed URLs with 1-hour expiry from Cloudflare R2.
- **Admin Authentication:** Admin panel requires email + password + role-based access control (RBAC). Super Admin and Content Admin roles.
- **Dependency Security:** Automated vulnerability scanning via GitHub Dependabot; npm audit in CI pipeline.

9. UI/UX Design Guidelines

9.1 Design Philosophy

The app must evoke calm and focus. Every design decision should reduce visual noise and support the meditative experience. We follow the principle of 'calm technology' - the interface should be present when needed and invisible when not. This contrasts with typical engagement-driven app design; our goal is to support habit formation, not maximize screen time.

9.2 Visual Design System

Element	Specification
Primary Color	#1B4332 (Deep forest green) - trust, nature, wellness
Secondary Color	#2D6A4F (Medium green) - growth, balance
Accent Color	#40916C (Soft green) - CTAs, highlights
Background	#FAFAF5 (Warm off-white) - warmth, not clinical
Surface	#FFFFFF (Pure white) - cards, modals
Text Primary	#1A1A2E (Near black) - readability
Text Secondary	#6B7280 (Gray) - labels, captions
Typography	Inter (body), Playfair Display (headings, quotes)
Spacing System	4px base unit (4, 8, 12, 16, 20, 24, 32, 48, 64)
Border Radius	12px cards, 8px buttons, 24px pills, full-round avatars
Shadows	Soft, minimal - only on elevated surfaces (cards, modals)
Animations	Gentle easing (300ms), breathing animations on meditation screen

9.3 Key Screen Layouts

- **Onboarding:** 3 screens with illustrations, minimal text, large CTA buttons. Skip option available.
- **Home:** Scrollable feed with daily quote card, streak widget, content carousels, and bottom tab navigation.
- **Meditation Timer:** Full-screen, dark/muted background, large timer display, ambient sound selector, minimal controls.
- **Course Player:** Video/audio player with lesson list drawer, progress indicator, notes section.
- **Profile:** Stats summary at top, settings list below, clean segmented layout.

9.4 Accessibility

- WCAG 2.1 AA compliance for color contrast ratios (minimum 4.5:1 for normal text).
- Screen reader support with proper accessibility labels on all interactive elements.
- Dynamic type support - text scales with system font size preferences.

- Reduced motion mode - disables animations for users with vestibular sensitivities.

10. Development Phases & Sprint Plan

The project is divided into 4 major phases across approximately 16-20 weeks. Each phase contains 2-week sprints with defined deliverables and review checkpoints.

Phase 1: Foundation (Weeks 1-4)

Sprint	Duration	Deliverables
Sprint 1	Week 1-2	Project setup (React Native CLI, Supabase, GitHub); DB schema design & migration; Auth flow (OTP login/register); Basic navigation structure with React Navigation; CI/CD pipeline setup (GitHub Actions + Fastlane)
Sprint 2	Week 3-4	Onboarding screens; Home screen UI with mock data; User profile CRUD; Admin panel scaffold (Vite + React); Supabase RLS policies implementation

Phase 2: Core Features (Weeks 5-10)

Sprint	Duration	Deliverables
Sprint 3	Week 5-6	Course listing and detail pages; Lesson player (video + audio); Course enrollment flow; Admin: Content upload and course management
Sprint 4	Week 7-8	Meditation timer with ambient sounds; Habit tracking & streak system; My Journey dashboard; Daily quote system; Admin: Quote management
Sprint 5	Week 9-10	Content directory with search; Events listing and registration; Push notification integration (FCM); Admin: Event management and notification dispatch

Phase 3: Monetization & Polish (Weeks 11-14)

Sprint	Duration	Deliverables
Sprint 6	Week 11-12	Payment integration (Razorpay); Subscription management; Premium content gating; Paywall screens; Admin: Subscription analytics
Sprint 7	Week 13-14	Live event streaming; Offline content downloads; Performance optimization; Sentry integration; Admin: Analytics dashboard (Mixpanel)

Phase 4: Testing & Launch (Weeks 15-18)

Sprint	Duration	Deliverables
Sprint 8	Week 15-16	Comprehensive QA testing; Bug fixes and UI polish; Beta testing with 50-100 users; App store listing preparation (screenshots, descriptions)
Sprint 9	Week 17-18	Final bug fixes from beta feedback; App Store & Play Store submission; Admin panel deployment to Vercel production; Launch monitoring setup

11. Testing Strategy

Test Type	Tool	Scope	Frequency
Unit Tests	Jest + React Native Testing Library	Utility functions, hooks, state logic, API handlers	Every PR
Component Tests	React Native Testing Library	Individual UI components render correctly	Every PR
Integration Tests	Jest + Supertest	API endpoint testing with DB interactions	Every PR
E2E Tests (Mobile)	Detox / Maestro	Critical user flows: login, meditation, enrollment, payment	Pre-release
E2E Tests (Admin)	Cypress	Admin panel workflows: content upload, user management	Pre-release
Performance Tests	React Native Perf Monitor + Lighthouse	App startup time, frame rate, API latency	Weekly
Security Tests	OWASP ZAP + npm audit	API vulnerability scanning, dependency audits	Weekly
Manual QA	Internal team + Beta testers	UX testing, edge cases, device compatibility	Sprint end

11.1 Quality Gates

- No PR can be merged without passing all unit and integration tests.
- Code coverage must remain above 70% for business logic modules.
- All critical user flows must pass E2E tests before any release build.
- Crash-free session rate must be above 99.5% before production release.
- App startup time (cold) must be under 3 seconds on a mid-range device.

12. Deployment & DevOps Pipeline

12.1 Environments

Environment	Purpose	Database	Hosting
Development	Local development and feature branches	Supabase (Dev project)	Metro bundler on device/simulator
Staging	QA and pre-release testing	Supabase (Staging project)	Fastlane internal distribution + Vercel preview
Production	Live user-facing app	Supabase (Production project)	App Store + Play Store + Vercel production

12.2 CI/CD Pipeline

The CI/CD pipeline is fully automated using GitHub Actions:

- **On Pull Request:** Lint (ESLint + Prettier), type-check (TypeScript), run unit + integration tests, build check.
- **On Merge to main:** Run full test suite, build staging apps via Fastlane, deploy admin panel to Vercel staging.
- **On Release Tag:** Build production apps via Fastlane, submit to App Store Connect and Google Play Console, deploy admin panel to Vercel production.
- **Hotfixes:** For JS-only changes, use CodePush for over-the-air updates bypassing app store review.

13. Monetization & Subscription Model

13.1 Pricing Tiers

Feature	Free	Premium Monthly	Premium Annual
Price	Free forever	INR 199/month	INR 1,499/year (save 37%)
Guided Meditations	10 introductory sessions	Full library (500+)	Full library (500+)
Courses	1 free course	All courses	All courses
Meditation Timer	Yes (basic sounds)	Yes (all sounds)	Yes (all sounds)
Streak Tracking	Yes	Yes + insights	Yes + insights
Daily Quotes	Yes	Yes	Yes
Live Events	View only	Full access + replay	Full access + replay
Offline Downloads	No	Yes (up to 25)	Yes (unlimited)
Content Directory	Limited browse	Full access	Full access
Ad-Free	Contains ads	Yes	Yes

13.2 Revenue Projections

Assuming a 5% trial-to-paid conversion rate in the first 6 months, increasing to 8% by month 12, with an average monthly subscription price of INR 166 (blended between monthly and annual): at 100,000 MAU with 8% conversion = 8,000 paid subscribers generating approximately INR 13.3 lakhs/month (approx. \$1,600/month) in recurring revenue.

14. Third-Party Integrations

Service	Provider	Purpose	Integration Method
Authentication	Supabase Auth + OpenAuth	OTP verification, JWT management, session handling	Supabase JS SDK
OTP Delivery	MSG91 / Twilio	SMS OTP delivery to Indian and international numbers	REST API from Edge Function
Payment Gateway	Razorpay	Indian payments: UPI, cards, wallets, net banking	Razorpay React Native SDK
Payment Gateway	Stripe	International card payments, Apple/Google Pay	Stripe React Native SDK
Push Notifications	Firebase Cloud Messaging	Push notifications for iOS and Android	@react-native-firebase/messaging
Analytics	Mixpanel	User behavior, funnels, retention, A/B testing	Mixpanel React Native SDK
Crash Reporting	Sentry	Real-time crash logs, breadcrumbs, performance traces	@sentry/react-native SDK
Video Streaming	Cloudflare Stream	Adaptive bitrate video delivery, DRM	HLS player + signed URLs
Media Storage	Cloudflare R2	Audio files, thumbnails, user avatars	S3-compatible SDK
Email (Admin)	Resend	Admin notifications, user welcome emails	REST API

15. Performance & Optimization

15.1 Mobile App Performance

- **App Size:** Target under 30MB initial download; lazy-load heavy assets.
- **Cold Start:** Under 3 seconds on mid-range Android devices (Snapdragon 600 series).
- **Frame Rate:** Maintain 60fps on scroll-heavy screens using FlatList with windowing.
- **Image Optimization:** Use WebP format, progressive loading, cached with react-native-fast-image.
- **Bundle Size:** Tree-shaking, code splitting with lazy imports, monitor with react-native-bundle-visualizer.
- **Background Audio:** Efficient audio session management to minimize battery drain; test on both platforms.

15.2 Backend Performance

- **Database Queries:** Indexed on all foreign keys and frequently filtered columns; use EXPLAIN ANALYZE for slow queries.
- **Caching:** Supabase built-in caching; React Query client-side caching with stale-while-revalidate pattern.
- **API Response Time:** P95 under 200ms for read endpoints, under 500ms for write endpoints.
- **CDN:** All static assets and media served via Cloudflare CDN with edge caching; minimal origin hits.
- **Connection Pooling:** Supabase manages PgBouncer for efficient database connection pooling.

16. Risk Assessment & Mitigation

Risk	Probability	Impact	Mitigation Strategy
App store rejection	Medium	High	Follow Apple/Google guidelines strictly; test in-app purchase flows early; submit for pre-review
Low user retention after Day 7	High	High	Implement smart notification timing; streak rewards; personalized content recommendations; A/B test onboarding
Payment gateway integration issues	Medium	High	Integrate Razorpay early in Phase 3; have Stripe as fallback; use test mode extensively
Content production bottleneck	High	Medium	Start content production in Phase 1; partner with existing meditation teachers; build a content pipeline with batch recording
Scope creep delaying launch	High	High	Strict MVP scope; feature freeze after Phase 2; post-MVP backlog for future sprints
Supabase service outage	Low	High	Monitor Supabase status; implement graceful degradation; cache critical data locally; have failover strategy
Team member unavailability	Medium	Medium	Cross-train team members; maintain documentation; use pair programming for critical modules
Poor audio/video streaming quality	Medium	High	Use Cloudflare Stream with adaptive bitrate; test on low-bandwidth connections; implement quality selector

17. Budget Estimation

17.1 Development Costs (18-week timeline)

Since this is a student project, the primary costs are infrastructure and services rather than developer salaries. Below are the estimated monthly operational costs post-launch.

Service	Free Tier Covers	Estimated Monthly Cost (Post-Free)	Annual Cost
Supabase (Pro Plan)	2 projects, 50K MAU	INR 2,000/month (\$25)	INR 24,000
Cloudflare Stream	First 1,000 min free	INR 400/month (\$5 per 1K min)	INR 4,800
Cloudflare R2	10GB free	INR 100/month (est.)	INR 1,200
Vercel (Pro)	Hobby plan free	INR 1,600/month (\$20)	INR 19,200
Sentry (Team)	5K events free	INR 2,000/month (\$26)	INR 24,000
Mixpanel	20M events free	Free (within limit)	Free
MSG91 (OTP)	N/A	INR 500/month (est. 5K OTPs)	INR 6,000
Apple Developer Account	N/A	INR 8,300/year (\$99)	INR 8,300
Google Play Console	N/A	INR 2,000 one-time (\$25)	INR 2,000
Domain & SSL	N/A	INR 100/month	INR 1,200
Figma (Education)	Free for students	Free	Free

TOTAL EST.

Development phase: mostly free tier usage. Post-launch monthly: ~INR 7,000-10,000/month (~\$85-120). Annual: ~INR 90,000-1,20,000 (~\$1,100-1,450).

18. Post-Launch Roadmap

18.1 Version 1.1 (Month 2-3 Post-Launch)

- Social features: share meditation streaks, invite friends, referral rewards.
- Sleep stories and bedtime meditation category with sleep timer.
- Apple Health and Google Fit integration for mindful minutes tracking.
- Multi-language support (Hindi, Marathi, Tamil, Telugu initially).

18.2 Version 1.2 (Month 4-5 Post-Launch)

- AI-powered personalized meditation recommendations based on usage patterns, mood logs, and preferences.
- Community forums and discussion groups moderated by instructors.
- Wearable integration (Apple Watch, WearOS) for guided breathing on-wrist.
- Corporate/Enterprise plans for workplace wellness programs.

18.3 Version 2.0 (Month 6-9 Post-Launch)

- Advanced analytics and wellness dashboard with mood trends, sleep correlation, and stress mapping.
- Teacher/Instructor marketplace: allow verified meditation teachers to publish content and earn revenue share.
- Group meditation sessions with synchronized timers and post-session sharing.
- Integration with smart speakers (Alexa, Google Home) for voice-guided meditation.
- Podcast creation tools for instructors within the admin panel.

End of Document

This document serves as the comprehensive blueprint for the MAM Spiritual Wellness & Meditation Application. It should be treated as a living document and updated as the project evolves through each development phase.

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